



△ **Opalized** chrysocolla cabochon

Chrysocolla

Specification

Chemical name Copper hydrosilicate | **Formula** $\text{Cu}_2\text{H}_2(\text{Si}_2\text{O}_5)(\text{OH})_4 \cdot n\text{H}_2\text{O}$ | **Colours** Blue, blue-green | **Structure** Orthorhombic
Hardness 2–4 | **SG** 2.0–2.4 | **RI** 1.46–1.57 | **Lustre** Vitreous to earthy | **Streak** Pale blue, tan, grey | **Locations** UK, Israel, Mexico, Czech Republic, Australia, DR Congo, USA

The term chrysocolla was first applied by the Greek philosopher Theophrastus in 315 BCE (see box, below) to various materials used in soldering gold, derived from the Greek *chrysos*, meaning “gold”, and *kolla*, meaning “glue”. Chrysocolla forms as a decomposition product of other copper minerals, mainly in arid regions, and is often intergrown with harder minerals such as quartz, chalcedony, or opal, yielding a more resilient gemstone variety. It is usually cut *en cabochon*, and translucent, richly blue-green chrysocolla is particularly prized.

Blocky surface



Uncut chrysocolla | Rough | This freshly broken chrysocolla rough has a granular outside surface, probably concealing fine gem material within.



Egg form | Carved | This ornament cut from chrysocolla stands 44.5cm (17½in) tall, potentially making it the world's largest egg carved from this material.

Finely detailed eye

Subtle colour variation

Carved berries



Bird ornament | Carved | Weighing 69 carats, this carving by Ronald Stevens of a bird resting on a cluster of berries is one of the all-time finest chrysocolla pieces.

Theophrastus

The man who first described chrysocolla

Theophrastus came to Athens at a young age and studied under Aristotle. He is often said to be the father of botany due to his work on plants, yet he carried out equally important studies of minerals, and his treatise *On Stones* was used by mineralogists until the Renaissance. He was the first scholar to attempt a systematic classification of gems and minerals and, although his work was superseded, he can be considered the forerunner of modern, scientific mineralogy.



Theophrastus A pupil of Aristotle, he was the first person in the ancient western world to write about minerals and rocks.